

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: CONTINUOUS AND BATCH COPOLYMERIZATION CONTROL

1. ABSTRACT

This report reviews patent evidence relevant to nitrile butadiene rubber (NBR) synthesis with emphasis on composition control and process routes that can support low-acrylonitrile grades. The single primary patent in the manifest, US5770660A, is directed to continuous free-radical emulsion copolymerization of acrylonitrile and butadiene, but its disclosed objective is actually high bound acrylonitrile NBR rather than low-ACN material. Even so, it is highly informative for understanding how monomer feed strategy, reactor staging, and conversion control govern acrylonitrile incorporation in NBR. The patent contrasts conventional batch polymerization, incremental batch addition, and a five-vessel continuous process, showing that staged butadiene addition can maintain a more consistent monomer ratio and improve tensile and environmental resistance properties.

The disclosed process charges acrylonitrile and butadiene to a first reactor in an approximately 80:20 ratio, then continuously transfers the reacting latex through multiple vessels while adding butadiene incrementally to preserve an unconverted monomer ratio of about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene. Example data show batch processes with 21/54, 37/54, 46/54, and 56/43 parts butadiene/acrylonitrile feed schemes, and a continuous five-reactor scheme using 56 parts acrylonitrile and 20 parts butadiene initially, followed by four 6-part butadiene additions. The resulting bound acrylonitrile contents range from 48.8 to 59.5 wt% depending on feed strategy and conversion.

Overall, the patent landscape here emphasizes emulsion polymerization, staged monomer addition, and composition control rather than low-ACN targeting. For low acrylonitrile NBR, the key transferable lesson is that acrylonitrile content is governed primarily by feed ratio and monomer depletion profile; therefore, lower ACN grades would be expected from deliberately reduced acrylonitrile feed, controlled conversion, and careful reactor staging to avoid runaway composition drift.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US5770660A
- Patent Title: Continuous polymerization process for producing NBR rubber having a high bound ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a continuous free-radical emulsion copolymerization process for acrylonitrile and butadiene in a series of reaction vessels. Acrylonitrile and an initial butadiene charge are fed to a first reactor, then additional butadiene is added incrementally in downstream reactors to maintain a controlled unconverted monomer ratio and produce NBR with more uniform composition and improved properties. - Initial monomer feed ratio: about 80 parts acrylonitrile to 20 parts $\pm 10\%$ butadiene — Claim 1 - Unconverted monomer ratio: about 70 to 90 parts acrylonitrile to about 30 to 10 parts butadiene — Claim 1 - Number of reaction vessels: 5 — Example VII - Acrylonitrile charge: 56 parts — Example V - Butadiene charge: 43 parts total — Example V - Initial butadiene charge: 22 parts — Example V - Additional butadiene additions: 8 parts, 7 parts, and 6 parts — Example V - Final conversion: 81.9 percent — Example V - Bound acrylonitrile content: 50.2 percent — Example V - Acrylonitrile charge: 56 parts — Example VI - Butadiene charge: 43 parts total — Example VI - Final conversion: 86.8 percent — Example VI - Bound acrylonitrile content: 50.6 percent — Example VI - Acrylonitrile charge: 56 parts — Example VII - Butadiene charge: 20 parts initially — Example VII - Additional butadiene additions: 6 parts to second, third, fourth, and fifth vessels — Example VII - Steady-state conversion in first vessel: 24 percent — Example VII - Steady-state conversion in second vessel: 40 percent — Example VII - Steady-state conversion in third vessel: 56 percent — Example VII - Steady-state conversion in fourth vessel: 72 percent — Example VII - Steady-state conversion in fifth vessel: 82 percent — Example VII - Final conversion: 82 percent — Example VII - Bound acrylonitrile content: 48.8 percent — Example VII - Acrylonitrile monomer charged: 60 wt. pct. — Table I - Butadiene monomer charged: 40 wt. pct. — Table I - Acrylonitrile bound in copolymer: 46 wt. pct. — Table I - Acrylonitrile monomer charged: 70 wt. pct. — Table I - Butadiene monomer charged: 30 wt. pct. — Table I - Acrylonitrile bound in copolymer: 48 wt. pct. — Table I - Acrylonitrile monomer charged: 80 wt. pct. — Table I - Butadiene monomer charged: 20 wt. pct. — Table I - Acrylonitrile bound in copolymer: 50 wt. pct. — Table I - Acrylonitrile monomer charged: 90 wt. pct. — Table I - Butadiene monomer charged: 10 wt. pct. — Table I - Acrylonitrile bound in copolymer: 53 wt. pct. — Table I - Acrylonitrile monomer charged: 100 wt. pct. — Table I - Butadiene monomer charged: 0 wt. pct. — Table I - Acrylonitrile bound in copolymer: 100 wt. pct. — Table I - Acrylonitrile monomer charged: 50 wt. pct. — Table I - Butadiene monomer charged: 50 wt. pct. — Table I - Acrylonitrile bound in copolymer: 43 wt. pct. — Table I - Acrylonitrile monomer charged: 40 wt. pct. — Table I - Butadiene monomer charged: 60 wt. pct. — Table I - Acrylonitrile bound in copolymer: 38 wt. pct. — Table I - Acrylonitrile monomer charged: 30 wt. pct. — Table I - Butadiene monomer charged:

70 wt. pct. — Table I - Acrylonitrile bound in copolymer: 32 wt. pct. — Table I - Acrylonitrile monomer charged: 20 wt. pct. — Table I - Butadiene monomer charged: 80 wt. pct. — Table I - Acrylonitrile bound in copolymer: 24 wt. pct. — Table I - Acrylonitrile monomer charged: 10 wt. pct. — Table I - Butadiene monomer charged: 90 wt. pct. — Table I - Acrylonitrile bound in copolymer: 15 wt. pct. — Table I - Acrylonitrile monomer charged: 0 wt. pct. — Table I - Butadiene monomer charged: 100 wt. pct. — Table I - Acrylonitrile bound in copolymer: 0 wt. pct. — Table I

Relevant Experimental Evidence - Example I: Conventional batch polymerization with 21 parts butadiene and 54 parts acrylonitrile gave 59.5 wt% bound acrylonitrile but only 50.2 percent conversion. - Example II: Two-increment batch addition improved conversion to 73.3 percent and produced 52.6 wt% bound acrylonitrile. - Example III: Charging acrylonitrile and butadiene as in Example II and extending polymerization gave 54.9 wt% bound acrylonitrile at 73.6 percent conversion. - Example IV: Three-increment butadiene addition reached 85.2 percent conversion and 50.5 wt% bound acrylonitrile. - Example VII: A five-vessel continuous process maintained staged conversions from 24 to 82 percent and produced 48.8 wt% bound acrylonitrile with superior tensile and environmental resistance versus batch processes.

Technical Relevance Although the patent targets high-acrylonitrile NBR, it is directly relevant because it shows how feed ratio and staged monomer addition control acrylonitrile incorporation in NBR. The same process logic can be inverted for low-ACN grades by lowering acrylonitrile feed and controlling conversion to prevent composition drift.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

US5770660A demonstrates that NBR composition is strongly governed by monomer feed ratio, staged butadiene addition, and reactor staging in emulsion polymerization. While the patent is aimed at high bound acrylonitrile NBR rather than low-ACN material, it provides a clear process framework for composition control: lower acrylonitrile content would be expected from reduced acrylonitrile feed, controlled incremental monomer addition, and maintenance of a stable unconverted monomer ratio across reactors. For low acrylonitrile NBR synthesis, the evidence supports emulsion copolymerization with careful feed management as the most transferable strategy.

5. REFERENCES

- US5770660A | Continuous polymerization process for producing NBR rubber having a high bound ... | Not disclosed | US | 1998 | <https://patents.google.com/patent/US5770660A/en>